

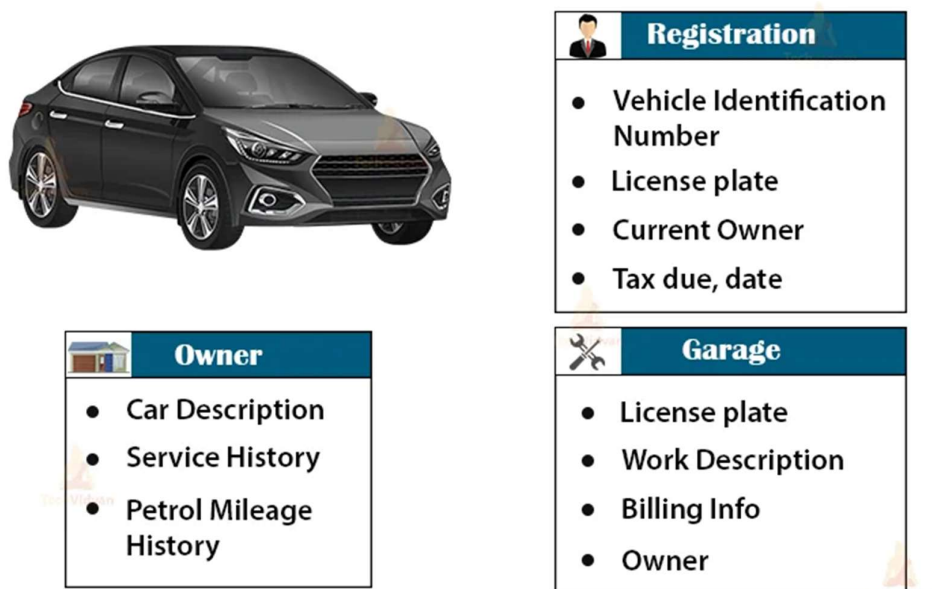
Abstraction in Java

+ What is Abstraction :-

- Hiding implementation details and showing necessary information.
- Hiding internal data and showing only functionality.

+ Real Time Example of Abstraction :-

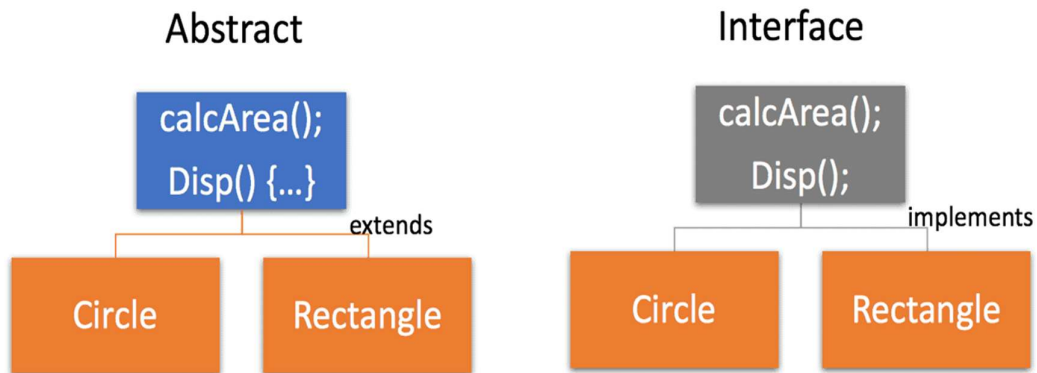
Example for Java Abstraction



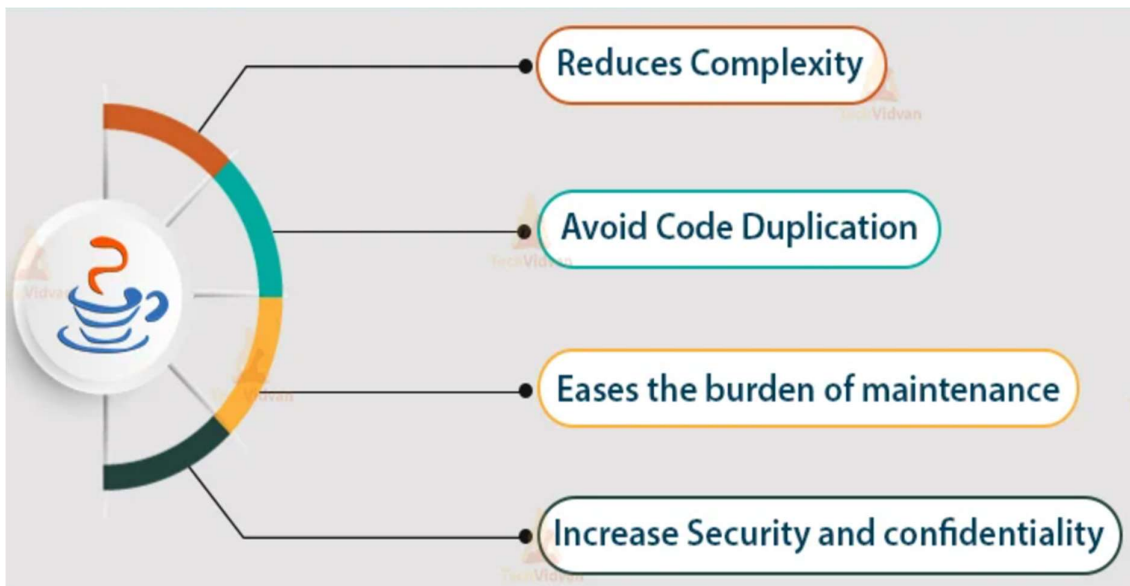
+ Key Points of Abstraction :-

- Achieved using **Abstract classes** and **Interfaces**.
- Helps in **reducing complexity** and **increasing code reusability**.
- You **cannot create objects** of abstract classes.
- Abstract methods **do not have a body**.

+ Types of Abstraction :-



+ Advantages of Abstraction :-



Difference between Abstract Class and Interface :-

FEATURE	ABSTRACT CLASS	INTERFACE
METHODS	Abstract & Concrete methods	Abstract methods (primarily), Default & Static methods (from Java 8)
VARIABLES	Instance variables (any access modifier)	Public, static, final fields (constants)
CONSTRUCTORS	Can have constructors	Cannot have constructors
INHERITANCE	Single inheritance (extends one class)	Multiple inheritance (implements multiple interfaces)
PURPOSE	Common base for related classes	Defines a contract for unrelated classes
IMPLEMENTATION	Can provide partial implementation	Defines a contract; implementation by implementing classes

Implementation of Abstraction

Input :-

```
1 package Oops;
2
3 public interface signal {
4
5     abstract void color();
6
7     default void location()
8     {
9         System.out.println("Pune");
10    }
11 }
1
1 package Oops;
2
3 public class signal_data implements signal {
4
5     @Override
6     public void color() {
7         // TODO Auto-generated method stub
8
9         System.out.println("Red color are stopping vehicle");
10        System.out.println("Green color ");
11        System.out.println("Yellow color");
12    }
13
14    public static void main(String[] args) {
15        signal_data ob=new signal_data();
16        ob.color();
17        ob.location();
18    }
19 }
20 }
```

Output :-

```
Red color are stopping vehicle
Green color
Yellow color
Pune
```